

Biological Human-Computer Interaction for Electronic Textiles

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Abstract

Electronic Textiles (E-textiles), also known as smart textiles, explore the intersection of electronics and textile fabrications, enabling sensing, actuation, and communication functionalities within textile forms. Although recent e-textile research has examined a wide range of fabrication methods and novel materials, the research heavily emphasizes the electronic sensing and actuation aspects yet lacks exploration of the involvement of biological systems. Recent advances in Biological Human-Computer Interaction (Bio-HCI) investigate the interplay between humans, computers, and biological systems, such as using biological materials as design elements or developing intermediate platforms to and from users. My research integrates Bio-HCI with e-textiles through (1) functional, (2) material, and (3) process investigations, focusing on three areas: biofluid sensing for personal healthcare, utilizing biodegradable materials for sustainable e-textiles, and biodesign for novel fabrication methods. My research advances scalable, accessible e-textile production, promoting ecological awareness and expanding the functional potential of e-textiles in wearable technology.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

E-Textiles, Wearable, Bio-HCI

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1 Introduction

Electronic Textiles (E-textiles), also known as smart textiles, explore the intersection of electronics and textile fabrications, enabling sensing [18, 30, 36] and actuation [11, 20, 27] functionalities within textile forms. E-textile practices often utilize common textile crafting techniques, such as weaving [10, 18, 23, 36, 38], knitting [20, 26], embroidery [14, 34], printing [21, 35], and braiding [29] to achieve seamless integration. E-textiles possess unique material and functional properties that interact with various biological systems,

including the human body, the surrounding environment, and other living organisms. However, recent e-textile research predominantly emphasizes electronic sensing and actuation aspects, neglecting the potential interactions with these biological systems.

Advancements in **Biological Human-Computer Interactions (Bio-HCI)** explore the relationship between humans, computers, and biological systems [31]. These developments open up new possibilities to incorporate various bio-based materials [2, 19], biosensing functions [43], and bio-participated fabrication processes [5, 28, 45] into human-computer interactions. By taking biological systems of the human body and other-than-human living organisms into consideration, these works present a holistic view situating human-computer interaction in a more-than-human-centered context [44].

My research aims to integrate Bio-HCI with e-textiles, creating new functionalities and fabrication techniques within textile forms. Textiles are soft and conformable to the skin, offering unique properties such as wicking and absorbing biofluids, making them an ideal interface for collecting and interacting with biological materials. Their comfort and cultural acceptance as wearable forms further enhance their usability. Furthermore, the widespread adoption of e-textile practices within the crafting and maker communities helps democratize fabrication techniques, expanding access to novel technologies for education and experimentation. My approach focuses on three key aspects: (1) **Functional exploration**: *Biofluid* sensing for personal healthcare. Functional exploration focuses on enhancing the functional capabilities of e-textiles beyond electrical and physiological sensing, valuing the user's human body as a biological system. Among these, I'm especially interested in biofluid sensing, turning human body "waste" into important personal healthcare resources. (2) **Material exploration**: *Biodegradable* materials for sustainable e-textiles. Material exploration emphasizes the materiality of e-textiles, specifically focusing on a sustainable e-textile prototyping approach using biodegradable materials addressing the end-of-life of e-textile swatches. (3) **Process exploration**: *Biodesign* for novel e-textile fabrication. Process exploration inquires how biodesign processes could be integrated into e-textile fabrication. I started with bio-inspired design and further explored how to integrate the growth of living organisms into the fabrication process.

I explored each of the approaches through a series of published and ongoing projects probing the possibilities of the integration of bio-HCI with e-textiles [46, 47, 49]. As e-textile practices are heavily rooted in the existing textile crafts, my research values the accessible fabrication of advanced materials that would enable a broader population to be involved and contribute. In my research, I synthesized design spaces and accessible fabrication approaches and conducted user studies to understand these new materials, functions, and processes. Furthermore, I emphasize the perspectives of textile practitioners as a core value of my work. Workshop

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studies have been carried out in several of my research projects to understand how textile practitioners would integrate the research outcome into their practices. Together, these work forms a biological e-textile design space that invites future explorations in this domain. My research goal is to leverage the integration to expand the impact of e-textile design and fabrication, promoting accessible personalization and ecological thinking.

2 Research Objectives and Methods

Through iterative prototyping and research through design [12, 50], I synthesize design methods and frameworks for integrating bio-HCI with e-textiles. Below are the two main research objectives:

(1) I examine textile crafts and craft-inspired digital fabrication methods [7] that provide accessible fabrication of e-textiles. Rooted in a material-centric approach, this hands-on method facilitates collaborative creation across disciplines. Through bio-HCI explorations, I identify *material*, *function*, and *processes* to provide technological resources for fabrication and design. I see this preliminary design space as an expandable framework where more textile crafts and Bio-HCI approaches can be included (Figure 1).

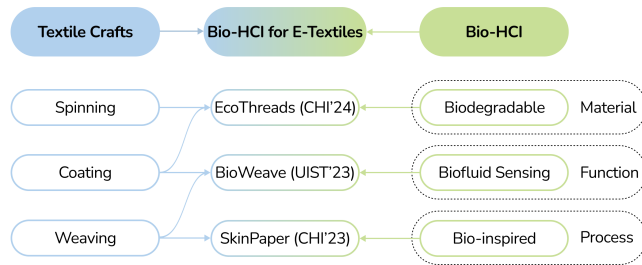


Figure 1: Preliminary Design Space

(2) I aim to understand how people work with and interact with bio-HCI-integrated e-textiles. Textile crafting embodies a distinctive creative process, which has been explored through various pedagogical methods including residencies [9], workshops [33], toolkits [6] and swatch exchanges [16]. My research emphasizes co-design and co-creation [37] through facilitating workshop studies to understand how textile practitioners integrate bio-HCI into their existing practices. Furthermore, through wearability studies [13, 24], I'm interested in examining how wearers experience novel interaction enabled through such e-textiles. I envision a co-design framework emphasizing the collaborative relationship between humans and materials to create e-textiles (Figure 2).

The anticipated contribution includes a framework and design space for the Bio-HCI for electronic textiles, with example projects demonstrating a variety of textile crafts and Bio-HCI approaches. Regarding the textile crafts aspect, I also expect to contribute to the accessible fabrication of advanced textiles, such as wet spinning, electrospinning, etc., through DIY fabrication tools and accessible materials. The technological contribution also includes toolkits or open-source fabrication instructions that artists, makers, and textile practitioners could follow and experiment with within their own practice.

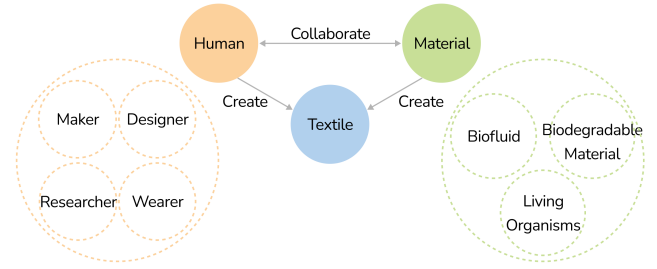


Figure 2: Preliminary Codesign Framework

3 BioWeave: Biofluid sensing for personal healthcare (UIST'23)

Sweating is a natural bodily function. While wearable sweat sensors have been thoroughly researched in the materials science and biochemistry fields, there has been comparatively less focus on wearable chemical sensors, as opposed to electrical or biometric sensors, in recent HCI research. By integrating biosensors with chemical sensing capabilities, we can facilitate the interaction with biological signals, particularly through bodily fluids [43], such as perspiration [39] and saliva [42]. However, textile sweat sensors that are conformable and natural to be worn during sweating activities are underexplored.

To explore how we can integrate biofluid sensing into textiles, we developed BioWeave, a woven thread-based sweat-sensing on-skin interface [47]. We synthesized accessible fabrication approaches to fabricate colorimetric and electrochemical sweat-sensing threads that can be woven into two- and three-dimensional structures conforming to diverse body locations. The BioWeave Design Space consists of sensing approaches, biomarkers, weaving patterns, structures, and suitable body locations. The colorimetric sensing threads are fabricated by dyeing cotton threads with colorimetric assays, which can be harvested from test strips. The electrochemical sensing threads are fabricated through a multi-layer coating structure. This design space is extensible and can be adapted to a wide range of biomarkers, including electrolytes (such as sodium), organic compounds (such as lactate), metabolites (such as glucose), and pH levels.

To demonstrate the BioWeave design space's extensibility, 2D and 3D weaving structures are explored for various body locations to seamlessly integrate multiple sweat-sensing threads into soft wearable interfaces as shown in Figure 3. The body locations include under the eye, under the arm, forehead, behind the ear, between the chest, and around the toes. We also conducted a preliminary wearability study with three participants to gain insight into the device's comfort and feasibility. This work probes the nuanced interaction of visualizing biosignals as wearable interfaces. As the colorimetric woven patterns may appear as a pattern integral to the textile itself, biosensing information can be conveyed to the wearer themselves but be perceived as an *ambiguous social display* [17]. BioWeave integrates biofluid sensing functions into textiles, offering personalized and unobtrusive analysis of health conditions.

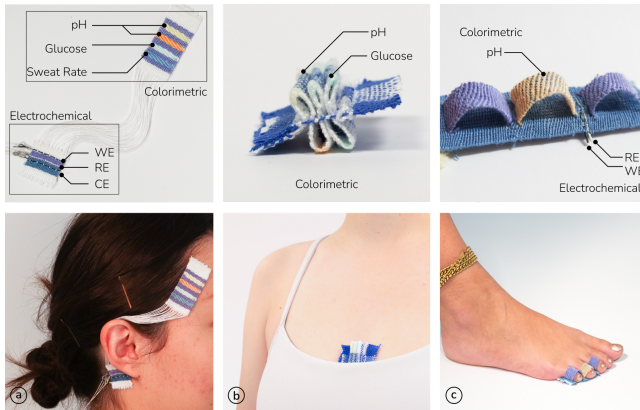


Figure 3: BioWeave woven textile-based sweat-sensing on-skin interface: (a) Forehead and ear glucose sensor with floating warp, (b) Between chest 3D pH sensor, (c) Between toes pH sensor.

4 EcoThreads: Biodegradable materials for sustainable e-textiles (CHI'24)

Integrating biodegradable materials promotes sustainable practices in fabrication processes, especially addressing the environmental impact of material waste during rapid prototyping. Previous studies have explored the use of biodegradable materials across various applications, including pottery [4], fiber [1], interactive interfaces [22], or 3D printing filaments [40]. I aim to identify compatible materials and develop corresponding fabrication approaches suitable for e-textile applications.

EcoThreads is a sustainable e-textile prototyping approach for fabricating biodegradable functional threads [49]. We synthesized two thread-based fabrication methods, wet spinning and thread coating, to fabricate functional threads from biomaterials or modify natural fiber to achieve conductive or interactive functionality. Within this design space, our emphasis lies on biodegradable threads with conductive and interactive properties, possessing the potential for sustainable seamless integration into textiles. We synthesized a sustainable e-textile life cycle prototyping workflow for individual creators using EcoThreads. We highlighted three cycles that intertwine with each other: the human perspective, the textile materials perspective, and the electronic component perspective. We understand that biodegradability is not a one-off solution towards sustainability. However, this offers opportunities that prior work cannot address, especially when reuse/recycle is not an option with the particular material choice or function to achieve.

To further explore the affordances of biodegradability and decaying as design elements, we ideated five transient e-textile use cases and prototyped them using the EcoThreads biodegradable functional materials (Figure 4). To understand how EcoThreads could be adapted to existing e-textile practices, we conducted a workshop study with five practitioners with various e-textile experiences. All five participants successfully incorporated EcoThreads with their specialized e-textile crafting techniques, ranging from embroidery and weaving to knitting, demonstrating EcoThreads' compatibility with existing e-textile practices. During the post-study interviews,

participants expressed their preference for fabricating their own functional threads using biodegradable materials and knowing the source of materials. This easy access also benefits the understanding of the environmental impact of the materials.



Figure 4: EcoThreads sustainable transient e-textile samples: (a) Braided capacitive touch sensing hair accessory, (b) Woven pH-sensing underwear, (c) Knitted cooling gel for fever.

5 SkinPaper: Biodesign for e-textile fabrication (CHI'23)

SkinPaper explores bio-inspired design process synthesizing functional waterproof materials for on-skin interfaces [46]. Inspired by the lotus effect, we render washi paper waterproof by adopting an accessible silicone coating method. We defined a twofold design space consisting of primitive elements and woven structures to create SkinPaper interfaces. The woven structure includes 2D Patterns, 2.5D Textures, and 3D Structures that are suitable for difficult-to-conform-to-body locations such as knees and shoulders that often pose a challenge for other on-skin substrate materials (Figure 5). Our eight-hour wearability study demonstrated SkinPaper's mechanical durability, water resistance, ability to maintain electrical functionality, and comfort when worn on the skin. Furthermore, we conducted a design workshop study with four participants to understand end-users' iterative design process from low- to high-fidelity prototyping using functional paper materials.

The aforementioned bio-inspired design approaches treat biological systems as passive elements in the design process—used as inspiration, resources, or targets for integration. However, an alternative approach, bio-participated design, positions living organisms as active collaborators in the creation process. Such processes integrate living organisms into human-computer interactions [32]. Examples of such include 3D printing of living responsive materials [25], inkjet printing of natto cells as nano actuators [45], and growing materials such as mycelium [41] and bacteria cellulose [3, 28] for wearable applications. My ongoing research explores integrating living plants into textiles to examine the design opportunities of *living textiles*. Through designing and fabricating living textiles following their natural growth cycle, I aim to probe how makers and wearers interact with this plant-textile hybrid to negotiate uncertainty and temporality.

6 Dissertation Status and Long-Term Goals

I am currently a third-year PhD candidate at the Department of Human-Centred Design, Cornell University, advised by Dr. Cindy Hsin-Liu Kao. I completed my candidacy exam in May 2024 and plan to defend my dissertation in May 2026. The duration of the

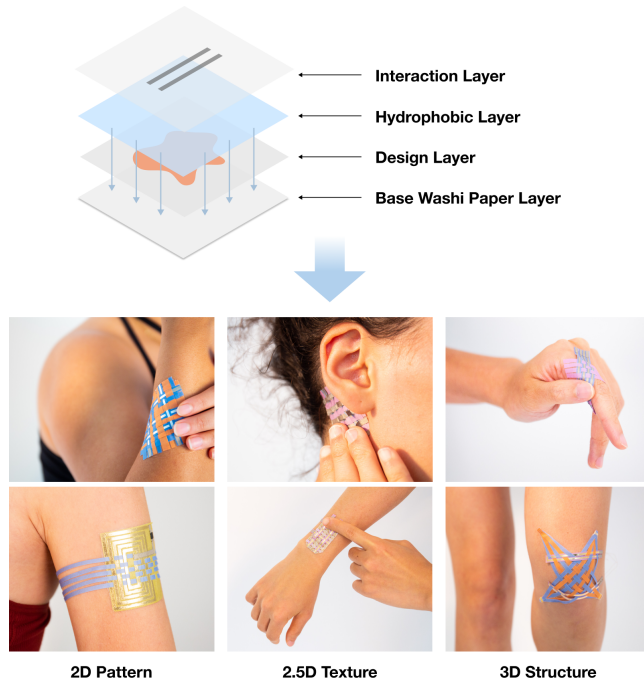


Figure 5: SkinPaper case studies spans from 2D patterns, 2.5D textures and 3D structures for on-skin interfaces.

PhD program in Human-Centered Design typically spans 4 years. I have not yet participated in a doctoral consortium. To date, I have published the aforementioned three full papers on my dissertation topic at CHI and UIST, and one ISWC short paper on scaling e-textile production [48], all as first author. Alongside my academic research in e-textiles, I also actively host community engagement activities such as sustainable e-textile workshops with textile practitioners and on-body artist residency, collaborating with makeup, tattoo, and nail artists. Some of the collaborations have also been showcased in academic conferences [8]. My long-term goal is to continue my multidisciplinary research at the intersection of biological human-computer interaction and electronic textiles. I aim to further expand this research by collaborating with practitioners from diverse textile communities and exploring the cultural dimensions of technology in craft.

7 Future Work

Currently, I'm working on several projects further extending the bio-HCI for e-textile design spaces, including energy harvesting from body fluids for self-powered wearables and integrating living organisms into growable textiles. I hope to contribute to the existing work of personalized healthcare interventions as well as sustainable design, to enable advanced material exploration with the approach of accessible textile fabrication.

The proposed interdisciplinary research spans multiple fields, such as fiber science, biochemistry, computer science, and design. Gaining subject-specific knowledge and skills is a challenge. To overcome this, establishing collaboration with researchers in related fields would be beneficial. Additionally, the lack of standard

evaluation methods and protocols for technical evaluation and user study in bio-HCI challenges the study design. I plan to combine the protocols for human trials from the material science field with research methods such as technology probes to synthesize suitable evaluation methods for the studies.

Furthermore, working with biofluid and biomaterials inevitably presents challenges to the unstable materiality, unique from common fabrication practices. Donna Haraway uses the term *sympoiesis* to address the making *with* materials in complex systems [15]. I'm interested in further exploring the collaborative creation between makers (such as textile practitioners) and materials (such as biofluid, biodegradable materials, and living organisms) to establish a code-sign framework seeing materials as entities (Figure 2). Lastly, I hope to explore more-than-human-centered approaches [44] to achieve ecological thinking of wearable technology and smart textiles.

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